GEO-03 Teacher Diagnostic Key

Teacher material - answers and diagnostics

Teacher Diagnostic Key

Use the diagnosis column to identify the missing bridge before assigning more practice.

ID | Answer | Independent check / diagnosis

GEO03-R01 | They are similar; congruence is not guaranteed. | AA establishes similarity; size is not fixed.

GEO03-R02 | Similar by SSS with scale 3/2; not congruent. | 9/6=12/8=15/10=3/2.

GEO03-R03 | 16:49. | Area ratio is the square of 4/7.

GEO03-R04 | 5:8. | Common altitude cancels in 1/2 bh.

GEO03-R05 | 2:1. | Centroid divides each median 2:1 from the vertex.

GEO03-R06 | They are equal. | The two subtriangles share the altitude from A and have equal bases.

GEO03-R07 | 2/5. | AA similarity gives corresponding scale AD/AB=DE/BC.

GEO03-R08 | 4:9. | Square the linear ratio.

GEO03-R09 | No. | SAS is unavailable without the included angle; SSS is unavailable without the third ratio.

GEO03-R10 | Both; the scale factor is 1. | SSS congruence; similarity scale is 1.

GEO03-R11 | 1/3. | Centroid is one-third of the altitude from the opposite vertex relative to each full side.

GEO03-R12 | No. | Equal area alone does not determine shape.

GEO03-P01 | 10 | AD/AB=6/9=2/3=AE/AC, so AE=10.

GEO03-P02 | 18 | Scale factor is 2, so 9 -> 18.

GEO03-P03 | 9:16 | (3/4)^2=9/16.

GEO03-P04 | 16 | Length scale=sqrt(144/81)=4/3; 12*(4/3)=16.

GEO03-P05 | 28 | Area ratio 13:7; 52*(7/13)=28.

GEO03-P06 | 28 | Area ratio 9:4; 63*(4/9)=28.

GEO03-P07 | 63 | A median gives equal areas.

GEO03-P08 | 50 | [GBC]=[ABC]/3.

GEO03-P09 | 84 | Six equal parts: 6*14=84.

GEO03-P10 | 4:25 | Linear scale 2/5, area scale 4/25.

GEO03-P11 | 32 | 200*(4/25)=32.

GEO03-P12 | 54 | A coordinate or ratio intersection gives M at one-quarter of the side above CD and centered horizontally, so height to AB is 9; area=12*9/2=54.

GEO03-P13 | 5 | The invariant ratio [GYZ]/[ABC]=1/48; 240/48=5.

GEO03-P14 | 45 | Same altitude from A to BC gives area ratio BP:PC=2:5; 18*(5/2)=45.

GEO03-P15 | 15 | AG=(2/3)AM, so AM=15.

GEO03-P16 | 150 | AD/AB=3/5; 54/[ABC]=9/25, so [ABC]=150.

GEO03-P17 | 80 | Small triangle scale=1/3, so its area is 90/9=10; subtract from 90.

GEO03-P18 | 60 | Perimeter is linear: 36*(5/3)=60.

GEO03-M01 | 30 | AD/AB=8/20=2/5=AE/AC, hence AC=12*(5/2)=30.

GEO03-M02 | 15 | Side ratio=sqrt(25/49)=5/7; 21*(5/7)=15.

GEO03-M03 | Similar only. | AAA/AA fixes shape but not size.

GEO03-M04 | 56 | Same altitude gives area ratio 4:7; 32*(7/4)=56.

GEO03-M05 | 12 and 6. | AG=2/3*18=12 and GM=1/3*18=6.

GEO03-M06 | 108 | Each side-with-centroid triangle is one-third: 324/3=108.

GEO03-M07 | 50 | Triangle CDE is similar to CBA with scale 1/2, so area factor 1/4; 200/4=50.

GEO03-M08 | 150 | Same trisection geometry gives [MAB]=3/8 of square area; 3/8*400=150.

GEO03-M09 | 20 | Use ratio 1/48; 960/48=20.

GEO03-M10 | 7:4 | Same altitude from A to BC, so area ratio equals base ratio.

GEO03-M11 | 200 | Area factor=(5/2)^2=25/4; 32*25/4=200.

GEO03-M12 | No. | The included angles differ, so SAS similarity fails.

Historical anchors

- IOQM-2024-Q12: independently recomputed 96; official HBCSE key 96.

- IOQM-2023-Q05: independently recomputed 10; embedded HBCSE-linked MTAI key 10.

Diagnostic contrasts

- Student says congruent from equal angles -> reteach similarity-versus-congruence boundary.

- Student uses k instead of k^2 for area -> reteach dimensional scaling.

- Student starts centroid coordinates for a pure area share -> retrieve centroid area invariant first.

- Student copies a ratio across a sketch without proving parallelism/similarity -> require a licensing first line.